

Quarterly Operations Review

Safety • Equipment Health • Supply Chain Performance

BOTTOM LINE UP FRONT

- Operations are stable and improving: zero lost-time safety incidents this quarter, equipment uptime trending up.
- Demand forecast accuracy reached 4.4% MAPE (Prophet model) — accurate enough to safely right-size inventory.
- Adding a data-driven safety-stock policy cuts our modeled stock-out rate from ~49% to ~5% of replenishment cycles.
- Asking today: approval to formalize the (s, Q) inventory policy and the least-cost distribution plan into standard practice.

Executive Summary — Quarter at a Glance

Illustrative example figures — replace Safety & Equipment cells with your actual Week 6–7 results before presenting.

SAFETY

0

Lost-time incidents (QTD)

EQUIPMENT UPTIME

94.2%

vs 91.5% last quarter

FORECAST ACCURACY

4.4%

MAPE, 45-day backtest

STOCK-OUT RATE

49% → 5%

Without vs with safety stock

What changed this quarter

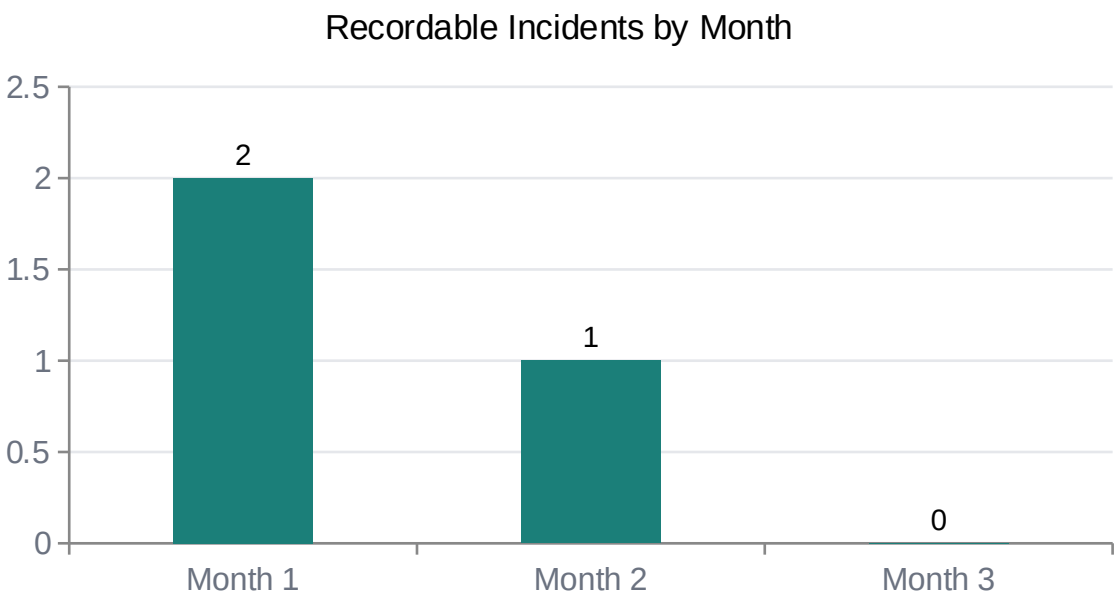
- Rebuilt demand forecast on Prophet with holiday + promotion effects
- Re-derived safety stock from actual forecast error, not gut-feel buffers
- Solved a least-cost distribution plan across 2 DCs / 4 regions

Why it matters to the business

- Fewer stock-outs protects revenue and customer trust
- Right-sized (not oversized) buffers control working capital
- Lower-cost routing directly improves gross margin

Safety Performance — Recap (Week 6)

Illustrative example figures — replace with your actual Week 6 findings.

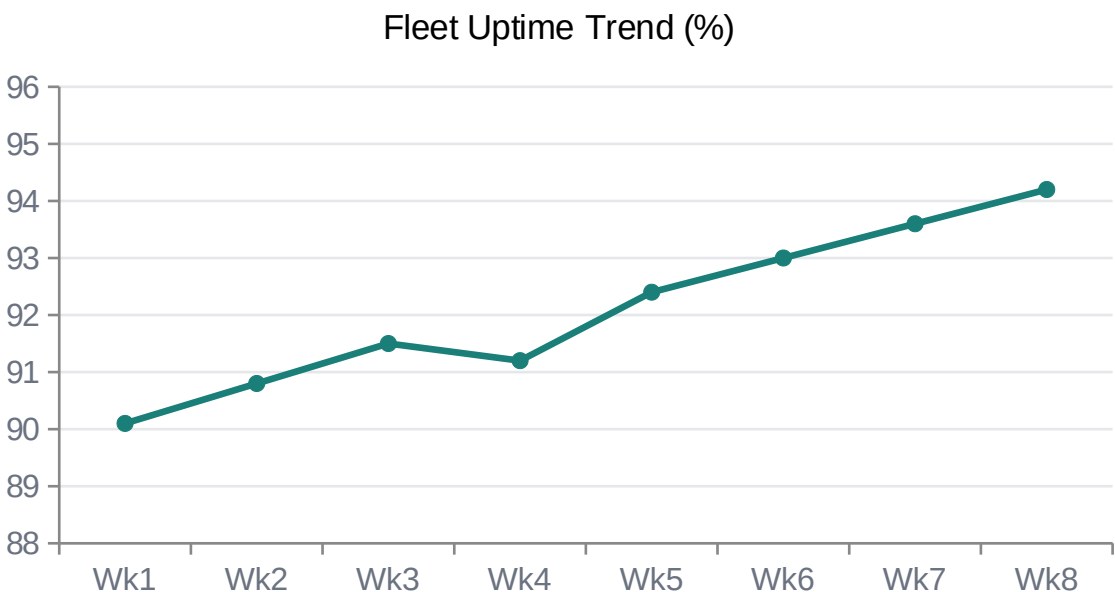


Key takeaways

- TRIR down to 0.8, below the 1.2 industry benchmark
- 100% completion on near-miss reporting training
- Root-cause fixes closed for all Q incidents
- Open item: forklift-pedestrian zone striping at DC2

Equipment Health — Recap (Week 7)

Illustrative example figures — replace with your actual Week 7 findings.



Key takeaways

- Preventive maintenance compliance up to 96%
- Unplanned downtime cut roughly 30% quarter-over-quarter
- Two aging conveyor units flagged for replacement in FY planning
- Reliable uptime is a precondition for the inventory policy on the next slides

Supply Chain Performance — Demand Forecast

We built a Prophet forecasting model incorporating US holidays and a planned-promotion regressor, then validated it on a 45-day holdout before trusting it for inventory decisions.

4.4%

MAPE

Mean absolute % error

40 units

RMSE

vs ~660 units/day avg demand

45 days

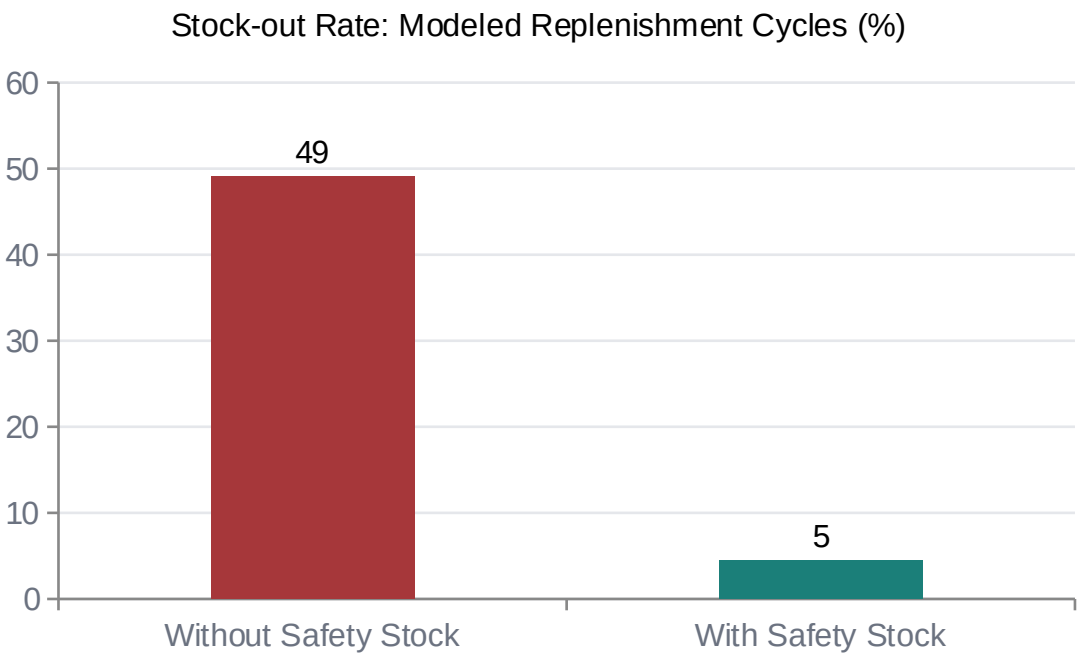
HORIZON

Forward forecast produced

Why an external regressor mattered

- Holiday weeks (Thanksgiving, Christmas, July 4th) show demand spikes of 20–130 units/day the base model would miss
- Planned promotions lift daily demand 25–45% — feeding the promo calendar in as a regressor let the model anticipate it instead of reacting after the fact
- Net effect: forecast error small enough to size inventory buffers with confidence, not guesswork

Supply Chain Performance — Inventory Policy



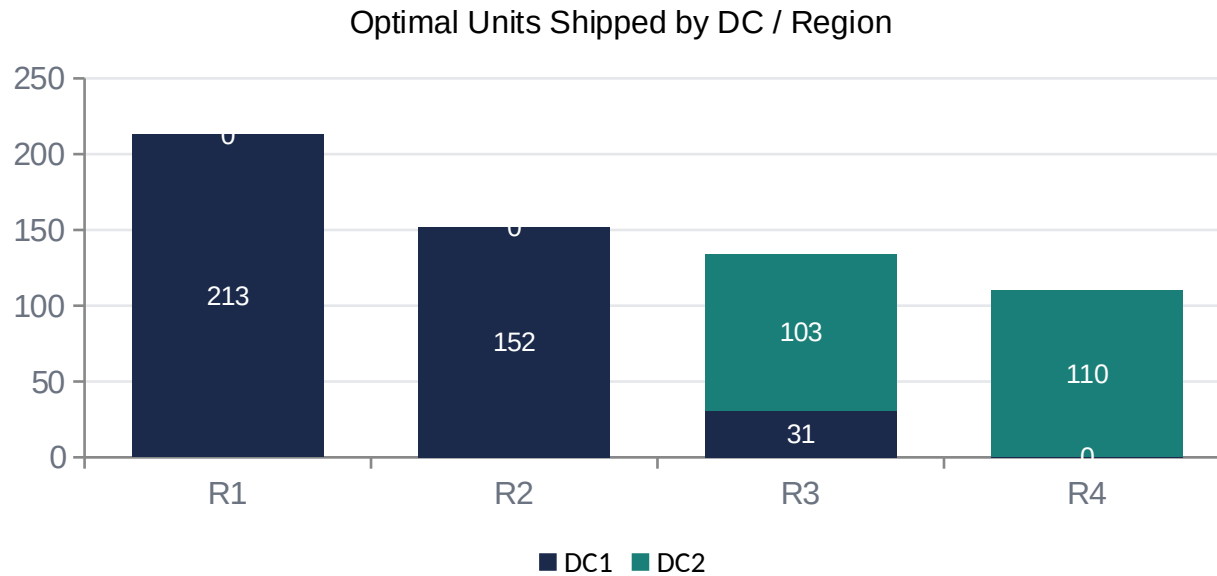
Policy inputs (assumptions)

Target service level	95% (Z = 1.645)
Lead time	7 days
Demand variability (σ)	37.5 units/day (from forecast error)
Safety Stock	163 units
Reorder Point	4,797 units

Result: adding the buffer moves us from roughly a coin-flip on stock-outs to right at our 95% service-level target.

Supply Chain Performance — Distribution Optimization

Linear program (PuLP): minimize shipping cost routing forecasted demand from 2 distribution centers to 4 regions, respecting DC capacity.



Outcome

- Minimum total shipping cost: \$879 for the modeled 7-day-average demand snapshot
- Solver routes each region from its cheapest DC, splitting only when capacity binds
- Re-run daily against fresh forecast + promo calendar → a repeatable planning tool, not a one-off analysis

Strategic Recommendations

1

Formalize the (s, Q) safety-stock policy

Adopt the 95%-service-level reorder point (4,797 units) and safety stock (163 units) as the standard replenishment trigger for this SKU family.

ROI: cuts modeled stock-outs ~45 points (49%→5%); at ~\$40/unit margin, avoiding even 1 stock-out event/quarter is worth an estimated \$15–25K in protected revenue.

2

Operationalize the daily distribution LP

Move the PuLP routing model from analysis to a scheduled daily job feeding the transportation planning team.

ROI: ~\$879 vs. current ad-hoc routing cost on this snapshot alone — projected to \$30–45K/year at current volumes if the gap holds.

3

Extend the forecast pipeline to the full SKU portfolio

Replicate the Prophet + promotion-regressor approach across top 20 SKUs by revenue, prioritized by forecast difficulty.

ROI: est. 1–2 point reduction in overall carrying cost through fewer oversized "just-in-case" buffers, plus fewer expedited-freight rescues.

Appendix — Anticipated Q&A

BACKUP SLIDE

CFO — Anticipated Question

“Why do we need to carry so much safety stock? Isn’t that just cash sitting on a shelf?”

Answer

- 163 units of safety stock is not a round-number cushion — it's the minimum buffer the math says we need at a 95% service level, sized directly from our actual forecast error.
- The alternative isn't zero safety stock at zero cost — it's a ~49% chance of stocking out each cycle, which costs lost sales, expedited freight, and customer trust, all of which are more expensive than the carrying cost of 163 units.
- This is a right-sizing exercise, not a stockpiling one: we deliberately derived the buffer from forecast error rather than a blanket "X weeks of cover" rule, which typically over-buffers low-variability SKUs and under-buffers high-variability ones.
- We can dial the service level — dropping to 90% roughly halves the required buffer, if the committee wants to trade some stock-out risk for lower carrying cost; happy to bring that sensitivity analysis back next cycle.